

Rational Numbers on the Number Line

Lesson 9-4

Name: Type your name **Date:** Today's date **Class:** Class period



TODAY'S OBJECTIVES

Content Objective: I can place rational numbers, including fractions and decimals, on a number line.

Language Objective: I can explain my reasoning using the words rational number, fraction, decimal, and number line.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Rational number

Fraction

Decimal

Number line

Equivalent

Integer



Tap any word to see what it means and a picture.

- 1 Any number that can be written as a fraction, including integers and decimals, is a

- 2 A number that shows part of a whole, like $\frac{3}{5}$, is a .

- 3 A number that uses a point to show parts of a whole is a .

- 4 A line with numbers in order used to place and compare values is a

- 5 Numbers that name the same value, like $\frac{1}{2}$ and 0.5, are

6 A whole number or its opposite is an .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Where would you place $-1\frac{1}{2}$ on the number line, and how do you decide between which two integers it falls?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Rational number** — Any number you can write as a fraction. This includes fractions, decimals, and integers.

Número racional — Cualquier número que puedes escribir como fracción. Incluye fracciones, decimales y enteros.

- ☐ **Fraction** — A number that shows part of a whole, like $\frac{3}{4}$.

Fracción — Un número que muestra una parte de un todo, como $\frac{3}{4}$.

- ☐ **Number line** — A straight line with numbers spaced out evenly.

Recta numérica — Una línea recta con números separados de forma pareja.

- ☐ **Integer** — Whole numbers and their opposites, like -2, -1, 0, 1, 2.

Número entero — Números enteros y sus opuestos, como -2, -1, 0, 1, 2.

- ☐ **Decimal** — A number with a dot, like 0.5, that shows a part less than one.

Decimal — Un número con un punto, como 0.5, que muestra una parte menor que uno.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

-1 $\frac{1}{2}$ is between ____ and ____.

-1 $\frac{1}{2}$ esta entre ____ y ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: All of these are rational numbers because each one can be ____.

Evidence: Student explains a rational number is any number that can be written as a fraction, including integers and decimals.

Reasoning: This evidence connects the definition of rational number to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **-1 1/2 is between ____ and ____.**

-1 1/2 esta entre ____ y ____.

- **It is closer to ____ because ____.**

Esta mas cerca de ____ porque ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.